

1 Introduction

This chapter presents the background and motivation of the Ethernacare project. It explains the social problem related to elderly individuals living alone and the risks linked to delayed discovery during emergencies. The chapter also outlines the project objectives, scope, and significance to provide a clear understanding of what the system aims to achieve.

1.1 Project Objective(s)

The first objective of Ethernacare is to provide an engaging safety check-in mechanism for older adults and other individuals who live alone. Users interact with a virtual companion named Oren to record a successful check-in. Instead of requiring a fixed daily interaction, the system allows the user to configure a rolling inactivity threshold between one and 168 hours. This approach provides flexibility for different user routines while still allowing the system to identify prolonged inactivity.

The second objective is to provide a staged inactivity response process. When the first inactivity window is missed, the application provides a local reminder to the user. When the second window is missed, the system attempts to send an SMS to the user's verified phone number. If the user remains inactive until the third window, the system escalates the SMS notification to the selected trusted contact. This process is intended to reduce unnecessary alerts while still providing an additional safety layer for users who may require assistance.

The third objective is to provide secure Legacy Planning functions. Users can record their funeral preferences, write protected Legacy Notes and upload private documents. The system also provides a server-controlled Legacy release process involving owner consent, a unique Legacy UID, a verified primary contact and protected access verification. If the user remains inactive for 90 days, the system warns the owner and provides 24 hours to check in or cancel the process before temporary access may be released to the verified primary contact.

The fourth objective is to improve regular user engagement through gamification. Oren has an energy level that changes according to inactivity and can be cared for through feeding and playing actions. Users can earn virtual tokens, purchase toys and select owned items for Oren. Check-in streaks separately allow users to unlock virtual badges and vouchers, providing motivation without requiring physical product delivery.

The fifth objective is to provide accessible general information through AI-assisted guidance. The AI feature helps users find basic information related to elderly care, emergency preparation

and funeral arrangements. However, its responses are provided for general informational purposes only and are not medical, legal, emergency or professional advice.

Overall, Ethernacare supports “SDG 3 – Good Health and Well-Being” by providing a practical smartphone-based safety-support application. The system does not diagnose medical conditions, confirm a user’s physical health, guarantee emergency rescue or guarantee successful SMS delivery. Its purpose is to reduce the possibility that prolonged inactivity remains unnoticed and to support communication between users and their trusted contacts.

1.2 Background of the Project

1.2.1 Malaysia is about to become an aging society

According to the prediction from “Portal Rasmi Kementerian Kewangan”, Malaysia will become an aged nation in 2048 (Azizan, 2025). The number of elderly people living alone will definitely increase as the elderly population increases. Moreover, many elderly people have poorer health compared to younger people and face a higher risk of death. These conditions increase the chance of dying alone without immediate discovery.

1.2.2 Migration and Elderly Parents Living Alone

In recent years, many young adults have moved from home to other areas for work or education. Internal migration remained dominant at 63.5% in 2024 (up from 62.3% in 2022), Urban to urban migration dominated at 84.6% in 2024 (up from 79.3% in 2022)(*Migration Survey Report, Malaysia, 2024, 2025*). This often happens when the children are grown up and want to start their own lives. As a result, parents are left living alone in their house. The contact and interaction between parents and children becomes less frequent, children are busy in their own business and usually only back to home on certain holidays. When elderly parents live alone, they are facing the higher risks of dying alone because help may not be available immediately. This situation increases social isolation.

1.2.3 Increase in the Number of Single Individuals

The number of marriages decreased 12.5 per cent from 215,022 (2022) to 188,100 (2023). (*Migration Survey Report, Malaysia, 2024, 2025*). In 2020, there were 14.5 male and 14.3 female births per thousand population, a decrease from 15.1 male and 14.9 female births per thousand population in 2019 (*International Journal Of Academic Research In Business & Social Sciences, 2024*). It was the lowest rate recorded in the country. Combined with these factors, the number of single individuals without children will increase in the future. Against the backdrop of soaring consumption levels, many people are facing delayed marriage, financial issues, work pressure or personal choices. Others remain single due to divorce or the loss of a spouse. As a result, more adults live alone for longer periods of time. This change affects daily support systems, especially as people grow older. Decades later, the number of living alone elderly people will inevitably increase.

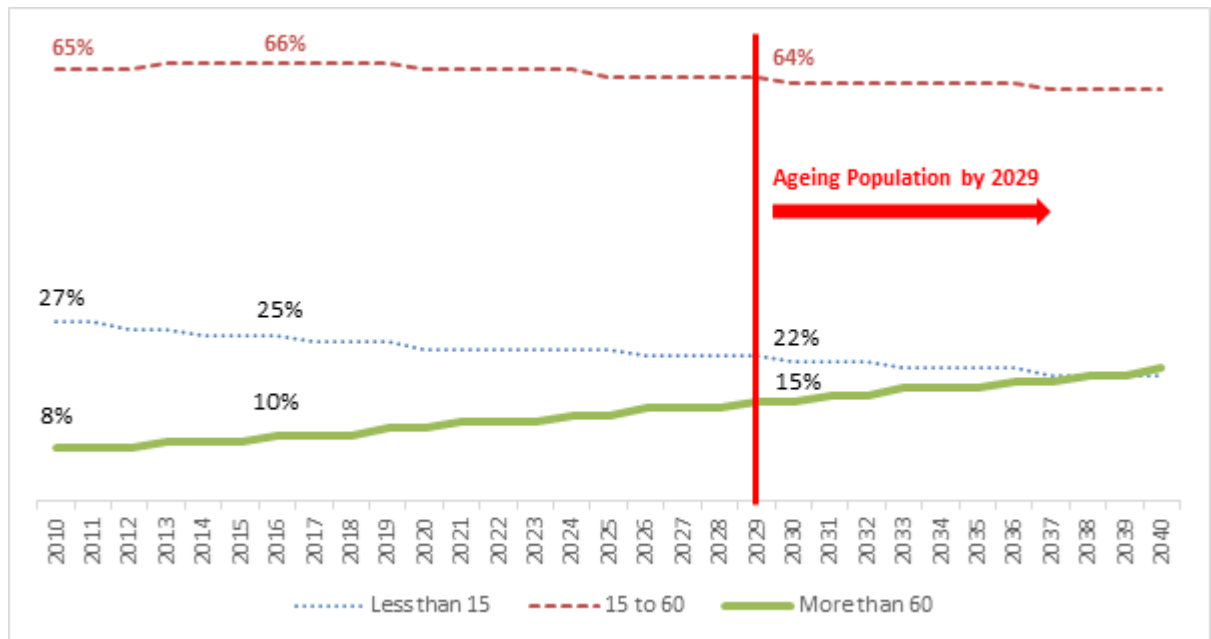


Figure 1.2.1: The prediction of the population age group in the future

(Malaysia Population Research Hub, n.d.)

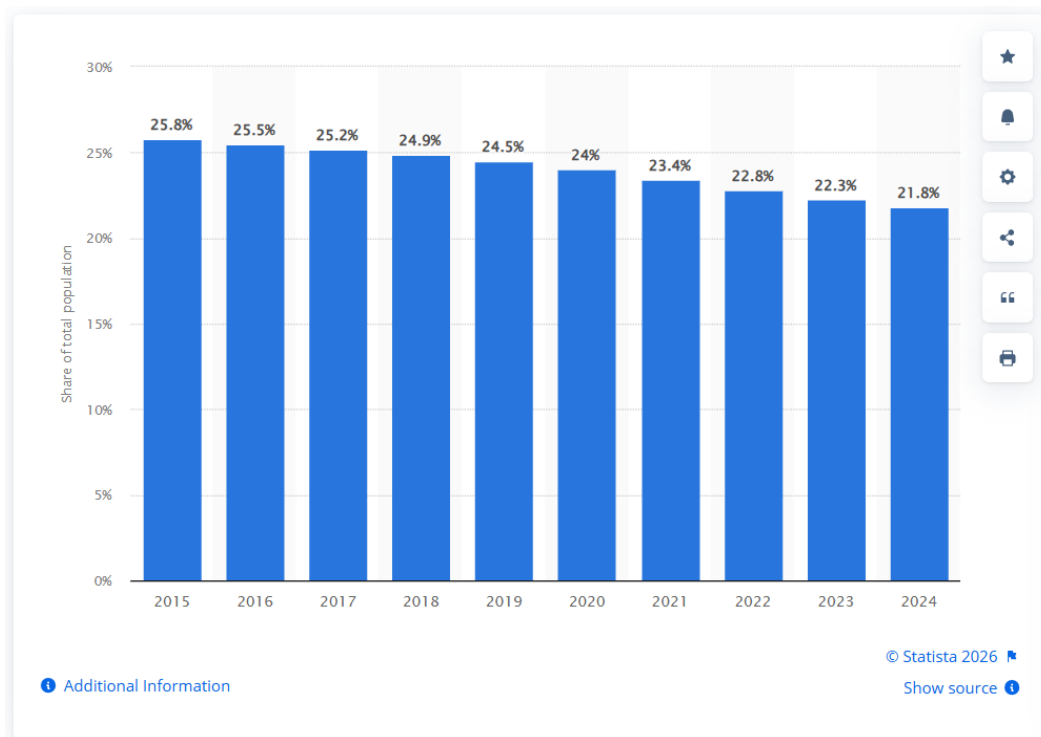


Figure 1.2.2: Children in total population in Malaysia from 2015 to 2024

(Statista, 2025)

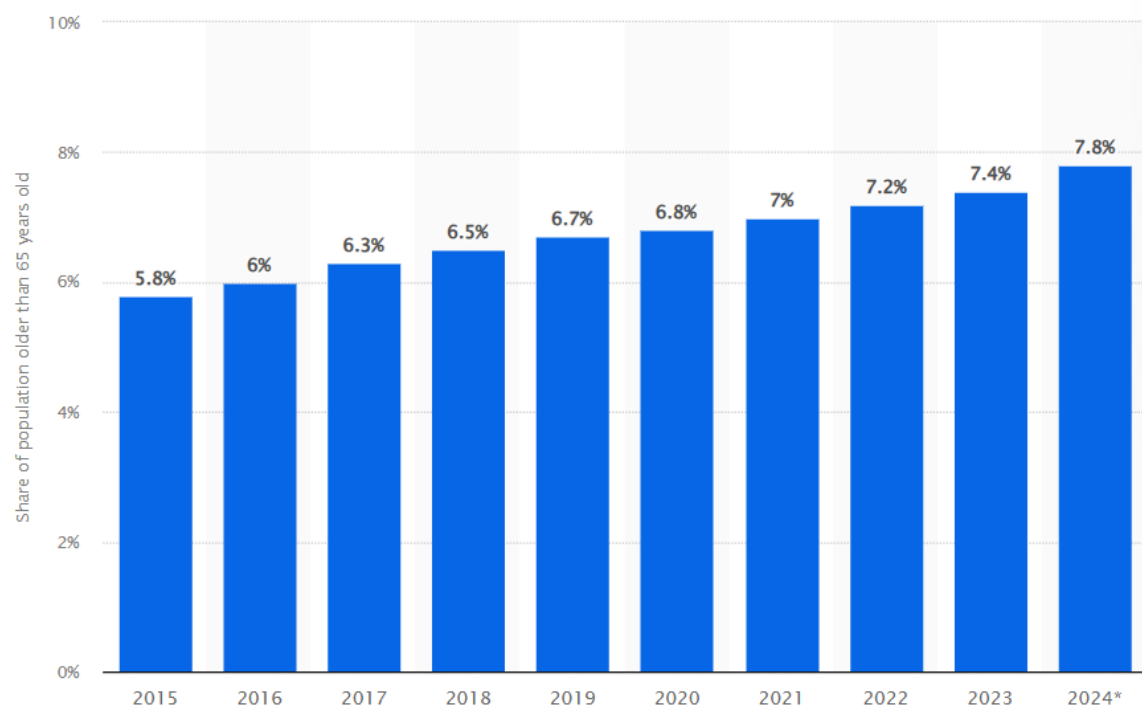


Figure 1.2.2: Aging population Malaysia 2015-2024

(Statista, 2025)